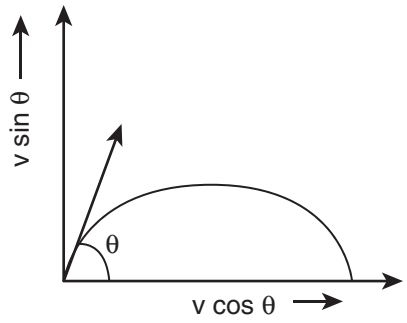
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5. A small particle of mass m is projected with an initial velocity v at an angle θ with x axis in X-Y plane as shown in Figure. Find the angular momentum of the particle.

Solution:



Let the particle of mass m cross a horizontal distance x in time t.

Angular momentum $\vec{L} = \int \vec{\tau}\, dt$

But $\vec{\tau} = \vec{r} \times \vec{F}$

$\vec{r} = x\hat{i} + y\hat{j}$ and $\vec{F} = -mg\hat{j}$

$$\therefore \vec{\tau} = \left(x\hat{i} + y\hat{j}\right) \times \left(-mg\hat{j}\right)$$

$$\vec{\tau} = -mgx\left(\hat{i} \times \hat{j}\right) = -mgx\hat{k}$$

$$\vec{L} = -mg\int (xdt)\hat{k} = -mgv\cos\theta\left(\int tdt\right)\hat{k}$$

Let initial time t = 0 and final time t = t_f

$$\therefore \vec{L} = -mgv\cos\theta\left(\int_0^{t_f} tdt\right)\hat{k} = -\frac{1}{2}mgv\cos\theta t^2{}_f\hat{k}$$

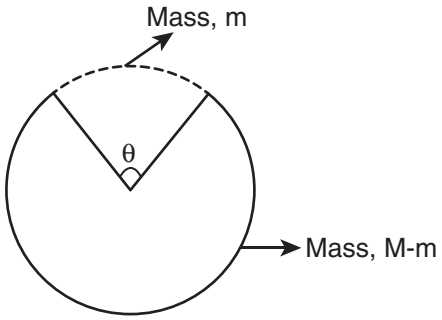
Negative sign indicates, $\vec{L}$ point inwards

6. From a complete ring of mass M and radius R, a sector angle θ is removed. What is the moment of inertia of the incomplete ring about axis passing through the center of the ring and perpendicular to the plane of the ring?

Solution:

Let R be the radius of the ring and M be the total mass of the complete ring.

Let m be the mass of the section removed from the ring then, mass of the incomplete ring is M-m



Let us introduce a positive integer (n), such that, $n\theta = 360^\circ$, or $n = \frac{360^\circ}{\theta}$

$$mass\ of\ incomplete\ ring = M - m$$

$$m = \frac{M}{360} \times \theta$$

$$\therefore mass\ of\ incomplete\ ring = M - \frac{M}{360} \times \theta$$

$$mass\ of\ incomplete\ ring = M - \frac{M}{n} = M\frac{(n-1)}{n}$$

For example, *a) when* $\theta = 60^\circ$; $n = \frac{360^\circ}{60^\circ} = 6$

$$\therefore n - 1 = 5$$

$$\text{mass of incomplete ring} = \frac{5}{6}M$$

b) when $\theta = 30^\circ$, $n = \frac{360^\circ}{30^\circ} = 12$

$$n - 1 = 11$$

$$\text{mass of incomplete ring} = \frac{11}{12}M$$